ELSEVIER

Precambrian Research 74 (1995) 225–244

Precambrian Research

Paleomagnetic investigation of the Neoproterozoic Gagwe lavas and Mbozi complex, Tanzania and the assembly of Gondwana

Joseph G. Meert [a], Rob van der Voo [b], Samwel Ayub [c]

[a] *Department of Geography and Geology, Indiana State University, Terre Haute, IN 47809, USA*
[b] *Department of Geological Sciences, University of Michigan, 1006 CC Little Bldg., Ann Arbor, MI 48109, USA*
[c] *Tanzania Division of Mines and Geology, Dodoma, Tanzania*

Received 16 August 1994; revised version accepted 27 February 1995

Abstract

The 810 Ma Gagwe–Kabuye lavas and the 743 Ma Mbozi gabbro-syenite complex of the Congo Craton in East Africa were sampled for paleomagnetic study in an effort to test a variety of tectonic models proposed for Neoproterozoic times. The paleomagnetic pole obtained from the Gagwe–Kabuye lavas falls at 25°S, 273°E ($\delta p = 7°$, $\delta m = 12°$) and compares favorably to a previously published paleomagnetic pole obtained from these rocks. The Mbozi complex pole yields a paleomagnetic pole at 46°N, 325°E ($\delta p = 5°$, $\delta m = 9°$) and differs significantly from a previously determined pole for the Mbozi complex. A comparison of these paleomagnetic poles to Laurentian poles of the same age suggests that the Congo Craton may not have constituted part of the Rodinia supercontinent in the configuration proposed by Dalziel (1992). An analysis of reliable paleomagnetic poles from the Gondwana blocks for the interval from 810 to 510 Ma reveals a coherent swathe of poles from 550 to 510 Ma and a scatter of pre-600 Ma poles. Our interpretation of the available paleomagnetic and tectonic data for this interval is consistent with the formation of Gondwana by two distinct orogenic events. This assembly resulted in the East Africa Orogen between 800 and 650 Ma and a younger Kuunga Orogen at 550 Ma outboard of the East Africa Orogen with possible sutures located in Sri Lanka, southern India and Enderby Land (East Antarctic Craton).

1. Introduction

The Meso–Neoproterozoic interval from 1100 to 550 Ma is receiving an ever increasing amount of attention following the seminal papers of Moores (1991), Dalziel (1991) and Hoffman (1991). These three papers outline the Meso–Neoproterozoic (1100–750 Ma) links between Australia, Antarctica and the western margin of Laurentia (called the SWEAT link), and the larger supercontinent of Rodinia. Rodinia consists of the separate cratonic blocks, i.e., the Siberian Craton, the Amazonia and Rio Plata cratons of South America, the Congo, West African and Kalahari cratons of Africa, Baltica, Australia, East Antarctica, India and Madagascar encircling the Laurentia continent (Fig. 1a). The configuration of Rodinia is based largely on the continuity of 1300–1100 Ma orogenic belts within the supercontinent (Fig. 1a). These 1300–1100 Ma orogenic belts are thought to represent a series of collisional events resulting in the formation of the supercontinent. The positioning of the Congo Craton (Africa) within Rodinia by Dalziel (1992) and Hoffman (1991) is based, in part, on the existence of roughly age-equivalent orogenic belts within the Congo Craton (called the Kibaran Belt, Fig. 1b) and the idea that the Congo and Kalahari cratons remained in close proximity throughout the Neoproterozoic (Hanson et al., 1993). However, Meert et al. (1994a)

0301-9268/95/$09.50 © 1995 Elsevier Science B.V. All rights reserved
SSDI 0301-9268(95)00012-7

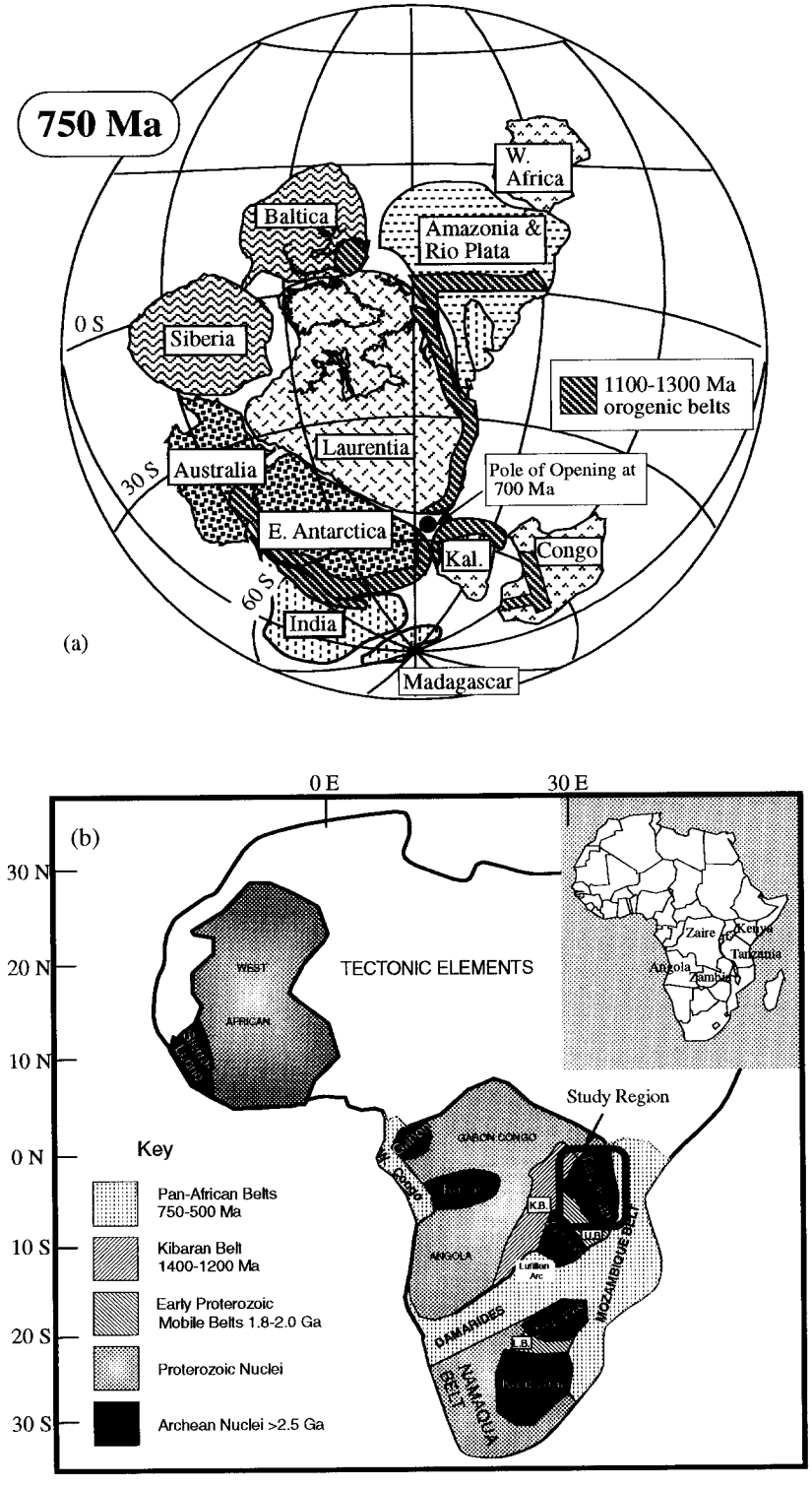


Fig. 1. (a) The supercontinent Rodinia reconstructed to its 750 Ma disposition according to Dalziel (1992). The 1300–1100 Ma orogenic belts are shaded and the presumed pole of opening between Laurentia and Australo–Antarctica is shown off the tip of the Kalahari Craton. (b) A regional map of Africa showing the outlines of the cratonic nuclei and the orogenic belts separating them (*K.B.* = Kibaran Belt; *L.B.* = Limpopo Belt; *U.B.* = Ubendian Belt). The study region is focused on the Tanzanian nucleus in East Africa.

demonstrate, on the basis of a well-dated 1236 Ma paleomagnetic pole from late Kibaran intrusions in Burundi, that if the Congo Craton was part of Rodinia the orogenic event responsible for its incorporation was post-1200 Ma.

The Rodinia configuration, as proposed by Dalziel (1992), is considered valid for the time period from 1100 to 750 Ma. By 750 Ma, evidence for rifting within Rodinia suggests that the supercontinent is beginning to break apart. The working model for the breakup of Rodinia (Dalziel, 1992) begins with the rifting of East Antarctica and Australia from western Laurentia between 750 and 700 Ma. The proto-Pacific ocean opens between these two blocks and western Laurentia as the Mozambique Ocean closes between the Congo Craton and Australo–Antarctica. Breakup along the northwestern, northern and eastern margins of Laurentia began at about 625 Ma, or later, as Siberia, Baltica and the Amazonian and Rio Plata cratons separate to open the Iapetus Ocean.

The breakup of Rodinia is synchronous with the assembly of Gondwana although the sequence and timing of events leading to the final assembly of Gondwana are debated. The assembly of Gondwana was originally perceived as a large-scale collision between East (India, Antarctica and Australia) and West (Africa and South America) Gondwana along the Mozambique orogenic belt (Fig. 1b; Burke and Dewey, 1972). Dalziel (1992) proposed that Gondwana assembly preceded the rifting events in eastern Laurentia. This means that, according to his model a second, short-lived supercontinent exists at 570–550 Ma with Gondwana juxtaposed against the eastern margin of Laurentia. Stern (1994) and Abdelsam and Stern (1993) view the formation of Gondwana as beginning with arc collisions in the Arabian–Nubian Shield sector of Africa at 800 Ma, followed by a major continent–continent collision between the Congo Craton and East Gondwana between 700 and 650 Ma. Both the models of Dalziel (1992) and Stern (1994) treat West Gondwana as distinct cratonic blocks (e.g., Rio Plata, Amazonia, Congo and West African cratons) assembled during a series of orogenic events. East Gondwana has been treated as a coherent unit from at least 1300 Ma until its breakup in the Mesozoic, but the concept of a coherent East Gondwana is questioned by Kröner (1993) and Meert and Van der Voo (1994). They propose that the East Gondwana continent is assembled through polyphase accretion of cratonic blocks between 800 and 550 Ma. Just as Rodinia assembled around Laurentia, the assembly of Gondwana can be viewed as Congo-centric and therefore knowledge of the position of the Congo Craton during this interval is critical in order to resolve the history of the amalgamation of Gondwana and, by proxy, the breakup of Rodinia. With these issues in mind we sampled the Mbozi syenite complex in southern Tanzania (743 ± 30 Ma) and the Gagwe lavas of western Tanzania and eastern Burundi (810 ± 25 Ma) in an effort to constrain the position of the Congo Craton during this interval of supercontinental dispersal and assembly.

2. Previous work and geology

The Gagwe and Kabuye lavas are part of the Bukoban Supergroup that crops out in western Tanzania and eastern Burundi (Fig. 2). Most of the Bukoban Supergroup consists of shallow-water sedimentary rocks with the lavas comprising the upper part of the Bukoban sequence. The Gagwe lavas of Tanzania, their counterpart Kabuye lavas in Burundi and a series of mafic dykes in Tanzania and Burundi are dated between 815 and 800 Ma by a number of K–Ar determinations [see De Paepe et al. (1991) and Tack et al. (1992) for a summary of ages]. Despite questions regarding the interpretation of K–Ar age determinations, the remarkable consistency of these ages and the bracketing of the age of the Gagwe lavas by other radiometric studies (De Paepe et al., 1991) leads to the conclusion that extrusion age of these lavas at 810 ± 25 Ma is reliable. De Paepe et al. (1991) suggested that the most of the Bukoban Supergroup was deposited in a local basin during the interval 950–800 Ma. The tectonic setting of the Gagwe lavas is not clear. Chemical discrimination diagrams are compatible with a convergent plate setting for the Gagwe and Kabuye lavas (De Paepe et al., 1991) but the authors noted that the data are scattered and offered as their preferred interpretation that the lavas erupted as continental flood basalts. The entire Bukoban Supergroup is only mildly deformed except near areas that were reactivated during Cenozoic extension (De Paepe et al., 1991; Tack et al., 1992).

The Gagwe lavas and the correlative Bukoban Intrusives of Tanzania were the subject of a previous paleomagnetic analysis (Piper, 1972) that yielded

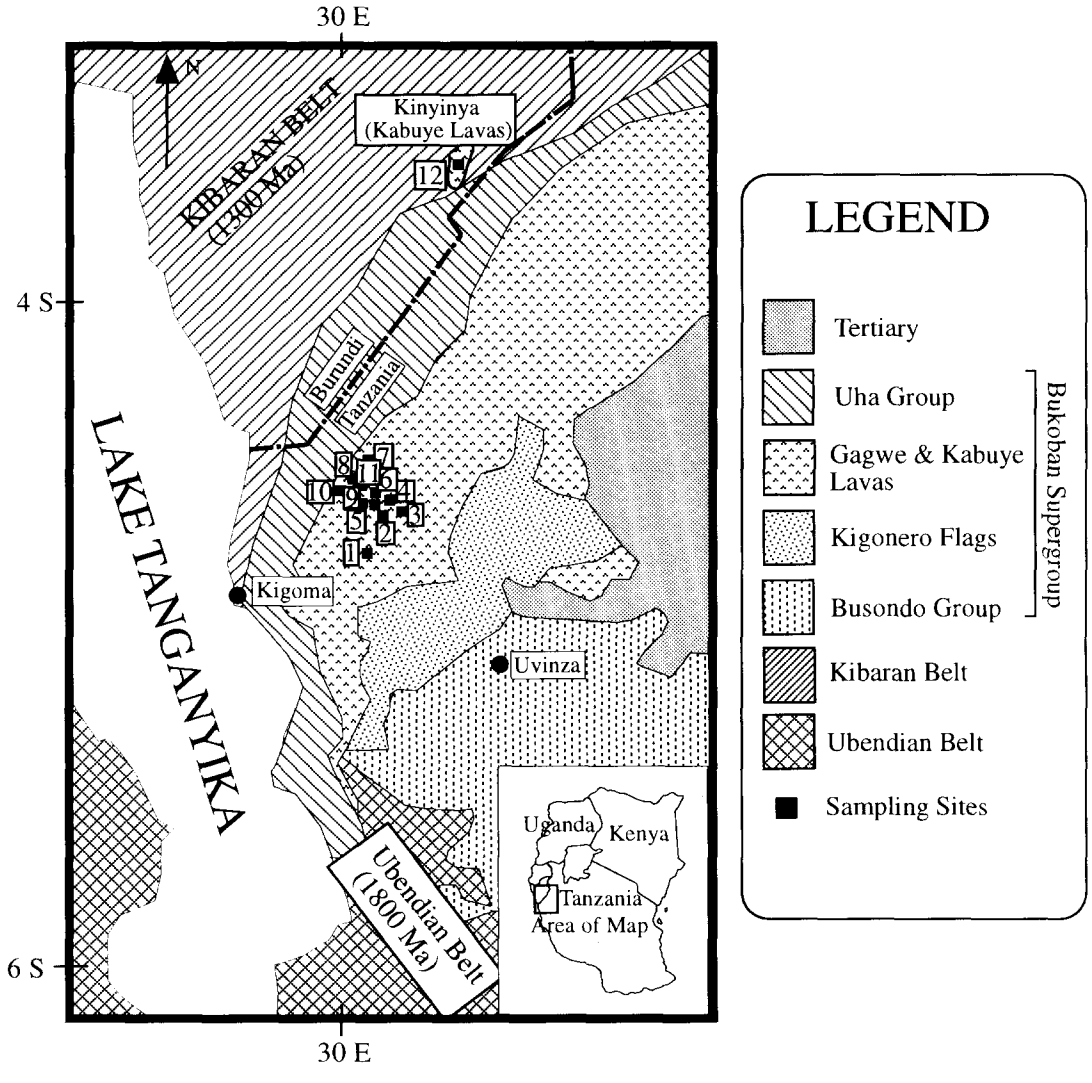


Fig. 2. Geologic sketch map of western Tanzania and eastern Burundi showing the locations of paleomagnetic sampling sites in the Gagwe and Kabuye lavas.

paleomagnetic poles at 29°S, 283°E ($k=5$, $\alpha_{95}=13°$) and 11°S, 281°E ($k=5$, $\alpha_{95}=19°$), respectively. Because the original paleomagnetic study was conducted without detailed demagnetization and rock magnetic tests, we resampled the Gagwe lavas at eleven sites in Tanzania and one site (Kinyinya) in Burundi (Fig. 2).

The Mbozi gabbro-syenite complex is a composite intrusive body covering approximately 300 km^2 of southern Tanzania (Fig. 3). The complex intrudes metasedimentary rocks of the Ubendian (1800 Ma) orogenic belt. The complex contains layered mafic rocks consisting primarily of gabbros and pyroxenite with sub-ordinate nepheline-syenite. Brock (1968) reports a K–Ar age of 743 ± 30 Ma obtained from a biotite sample in the syenitic rocks of the Mbozi complex. Mapping of the complex by Brock (1968) showed that the gabbros were faulted and intruded by

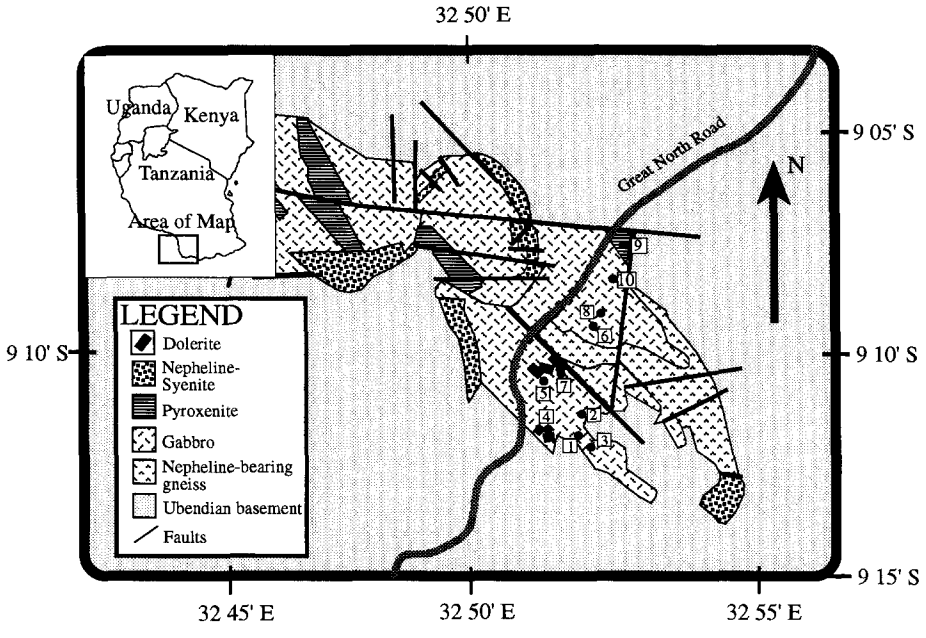


Fig. 3. Geologic sketch map of southern Tanzania and the Mbozi gabbro-syenite complex showing the locations of paleomagnetic sampling sites.

dolerite rocks before intrusion of the alkaline rocks. The Mbozi complex was the subject of a previous paleomagnetic study (Piper, 1975) that yielded a paleomagnetic pole at 68°N, 72°E ($k = 19$, $\alpha_{95} = 18°$). Individual site means of that study showed a very large scatter and the behavior of the samples during alternating field demagnetization was erratic. We resampled the Mbozi complex at ten sites in the gabbros, dolerites and pyroxenites (Fig. 3).

3. Paleomagnetic results

The locations of sampled sites are shown in Figs. 2 (Gagwe and Kabuye) and 3 (Mbozi complex). Five to thirteen cores were drilled from each site using a Pomeroy gasoline-powered drill and oriented with sun and magnetic compass. There was no significant deviation in orientation directions between the two methods. Two sites in the Gagwe lavas were also hand-sampled (two to three additional hand samples each) and cores were drilled in the laboratory at the University of Michigan. The cores were then cut to specimens ~22 mm in length for measurement using a 2-G Enterprises three-axis cryogenic magnetometer at the University of Michigan. Demagnetizations were carried out with an ASC-Scientific model TD48 thermal demagnetizer or a Sapphire Instruments alternating field demagnetizer.

3.1. Gagwe and Kabuye lavas

The intensity of the natural remanence magnetization (NRM) ranges between 10^{-5} and 4 A/m. Thermal and alternating field (AF) demagnetization on pilot specimens indicated that AF demagnetization resulted in a clearer definition of the magnetic components (see Figs. 4 and 5). Thermally treated samples often showed considerable decay of the remanence prior to reaching a consistent stable direction (Fig. 4). Demagnetization results were plotted in orthogonal vector diagrams (Zijderveld, 1967) and linear trajectories were used to determine directions of magnetic components through the use of principal component analysis (Kirschvink, 1980; Torsvik, 1992). Many samples showed great-circle trajectories toward a stable endpoint direction

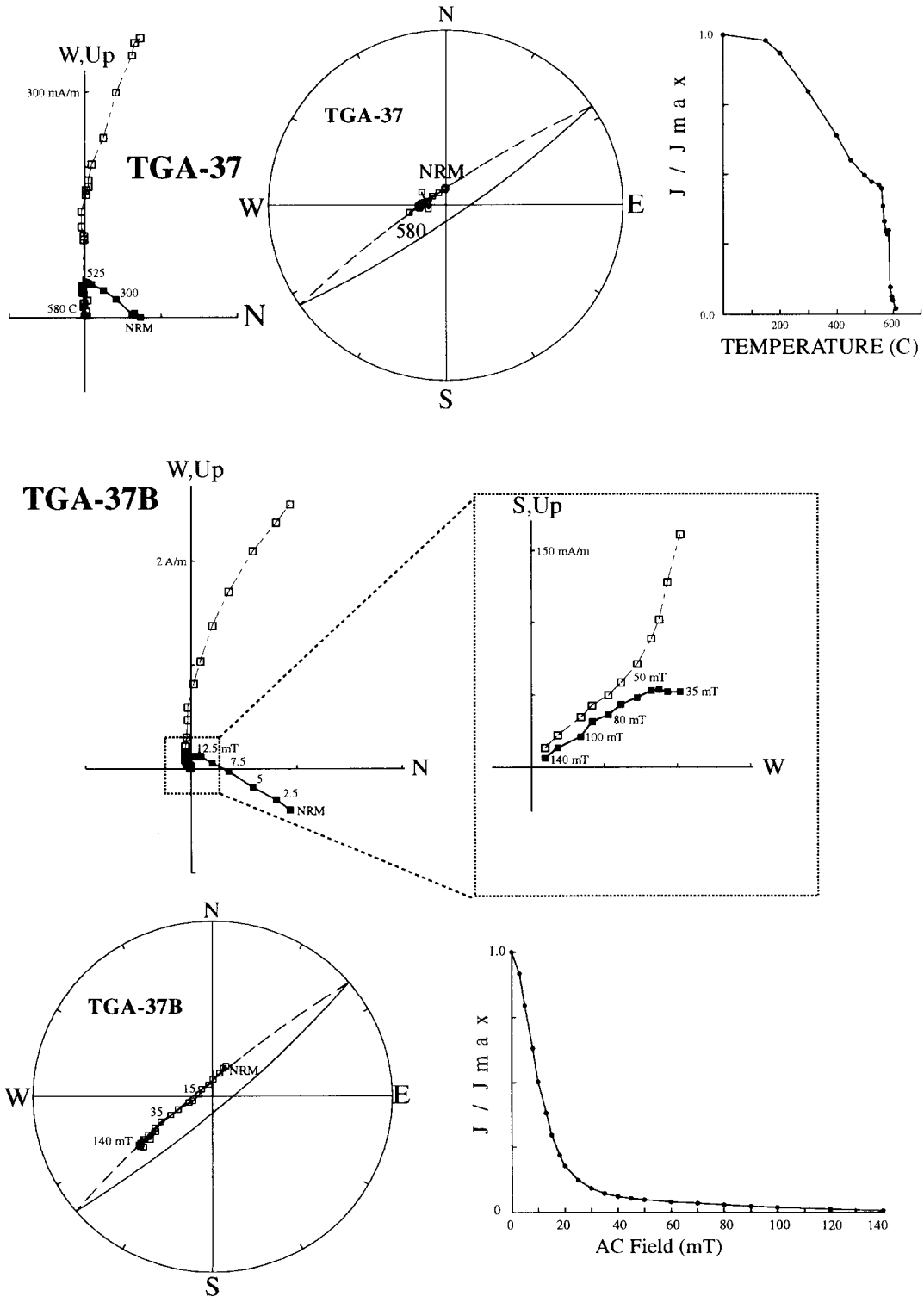


Fig. 4. Orthogonal vector diagrams, equal-angle stereoplots and intensity decay plots for pilot samples of the Gagwe lavas demonstrating the effectiveness of alternating field demagnetization over thermal demagnetization. In the orthogonal vector diagrams, open squares represent vectors projected onto the vertical plane and closed squares represent vectors projected onto the horizontal plane. In the stereoplots, circles represent projections onto the upper hemisphere and dots represent projections onto the lower hemisphere.

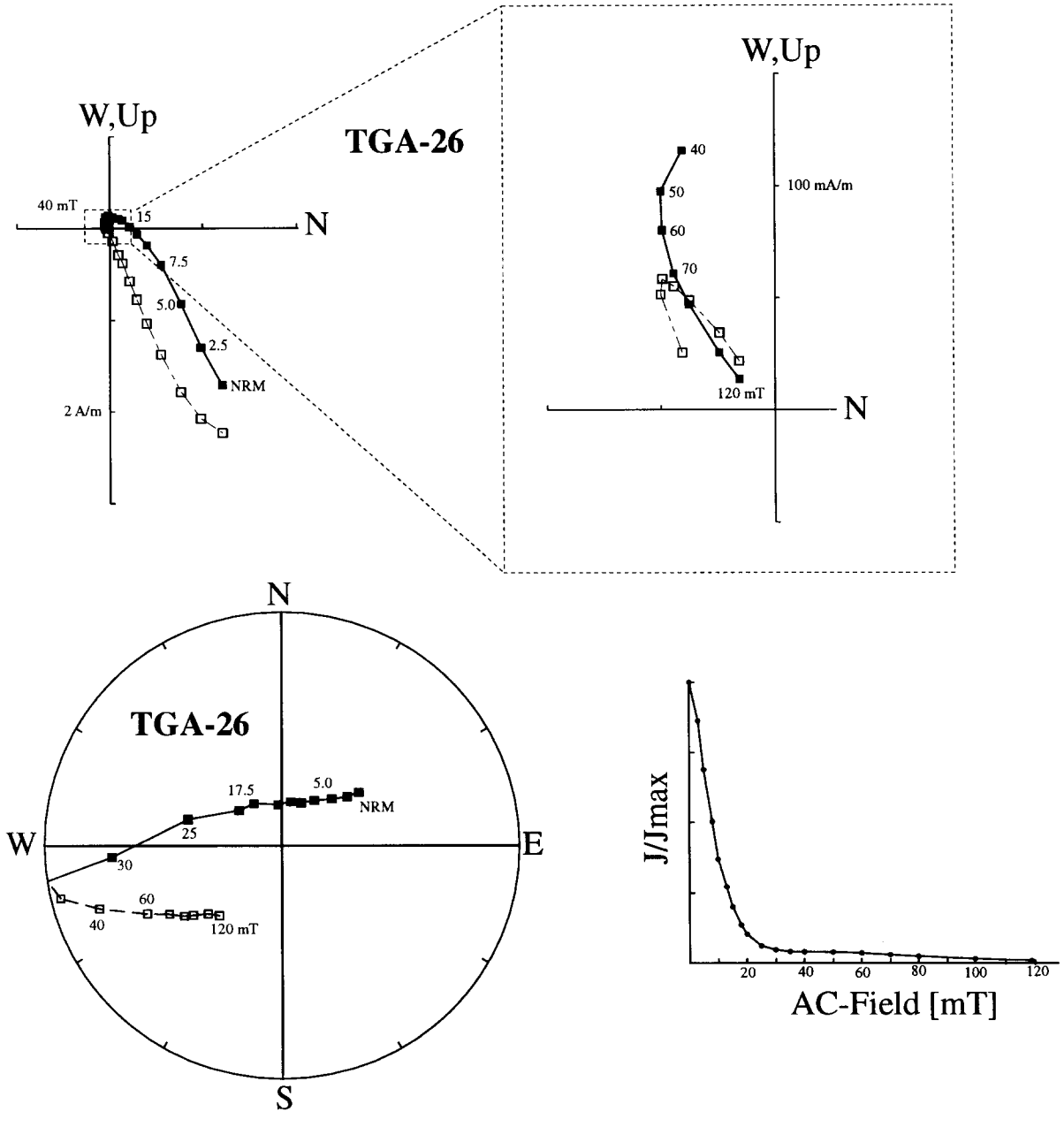


Fig. 5. Orthogonal vector diagram of Gagwe sample TGA-26 showing the great-circle trajectory towards the stable WSW intermediate-up direction. Symbols are the same as in Fig. 4.

that is easily resolved in the orthogonal vector diagrams (Figs. 4–6). The most common great-circle trend passes through a shallow WSW direction between 30 and 60 mT before reaching a stable WSW intermediate-up direction after demagnetization at higher fields (Figs. 5 and 6). Less common trajectories followed steeply up paths towards the WSW intermediate-up direction (Fig. 4). The first AF-removed low-coercivity components in the Gagwe lavas are random. These random low-coercivity components, coupled with the

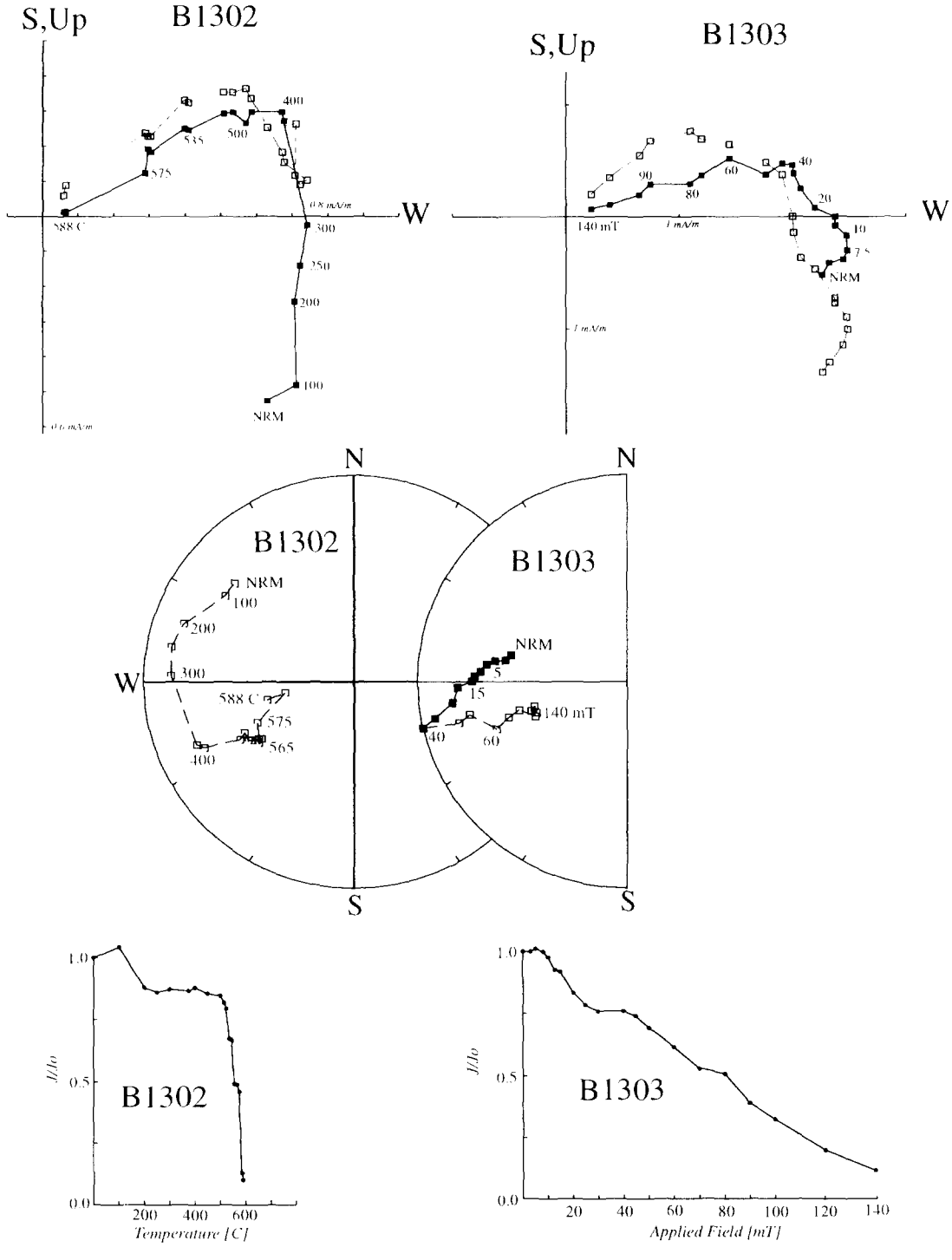


Fig. 6. Orthogonal vector diagrams of the Kabuye lavas in eastern Burundi showing sample behavior during thermal and alternating field demagnetization. Note the similar behavior and direction to the Gagwe lavas of Tanzania. Symbols are the same as in Fig. 4.

Table 1
Paleomagnetic results from the Gagwe and Kabuye lavas

Site	n/N	Dec. (°)	Inc. (°)	k	α_{95} (°)	VGP lat.	VGP long.
1-H	6/6	235	−27	31	12	31°S	280°E
2-H	6/7	247	−53	23	14	16°S	263°E
3-H	7/7	252	−07	49	09	19°S	295°E
4-H	10/10	245	−31	38	08	21°S	280°E
5-H	3/5	209	−74	126	11	26°S	221°E
6-H	8/8	239	−51	22	12	23°S	263°E
TC		243	−53				
7-H	7/7	228	−36	26	12	37°S	271°E
TC		230	−38				
8-H	6/7	255	−32	10	23	13°S	291°E
TC		257	−33				
9-H	7/8	245	−30	40	10	23°S	291°E
TC		246	−31				
10-H	11/11	245	−43	29	09	20°S	271°E
11-H	3/5	216	−43	54	17	44°S	258°E
12-H	6/6	241	−31	32	11	26°S	279°E
Total-H	12/12	240	−39	18	10	25°S	273°E
TC		241	−39	18	10	$\delta p = 7$	$\delta m = 12°$
BUK-I	16 sites	077	+32	8	19	11°N	101°E
Gagwe and intrusives	12 sites + Buk-I					25°S	274°E
						$K = 17$	$A_{95} = 10$

H = high-temperature or coercivity component; n/N = number samples used/number samples site; Dec. = declination (°); Inc. = inclination (°); k, K = Fisher precision parameter; α_{95}, A_{95} = 95% confidence cone (°); VGP lat. = latitude virtual geomagnetic pole; VGP long. = longitude virtual geomagnetic pole; δp, δm = semi-angles of 95% confidence oval about paleomagnetic pole in the co-latitude direction (δp) and at a right angle to the colatitude direction (δm); TC = tilt-corrected directions, other sites are flat lying. Note the final mean pole is based on the average of the VGP poles for the twelve sites in this study plus the mean (given unit weight) of sixteen sites in the Bukoban intrusives studied by Piper (1972). BUK-I = Bukoban intrusives mean result. Site 12 (3°S, 30.3°E) is the Kabuye lavas mean. Gagwe mean site latitude 4.5°S, 30.1°E.

general ineffectiveness of thermal treatment in isolating consistent site mean directions, is suggestive of a lightning-induced isothermal remanence overprint in the lavas.

A total of twelve sites from the Gagwe and Kabuye lavas yield a characteristic in-situ direction of $D = 240°$, $I = -39°$ ($k = 18$, $\alpha_{95} = 10°$). The sampled sites have tilt corrections that range from 0 to 4° and may well be primary dips; however, the tilt-corrected mean from all sites is identical to the in-situ mean with no change in the precision parameter kappa or α_{95}. Sites 3 and 5 yielded significantly shallower or steeper in-situ directions than the characteristic direction (Site 3 $D = 252°$, $I = -7°$; Site 5, $D = 209°$, $I = -74°$). This may be attributed to incomplete resolution of the characteristic remanence in those samples. Table 1 lists the individual site statistics and overall mean statistics and the results are depicted in Fig. 7. The paleomagnetic pole calculated from the site mean results is 25°S, 273°E ($\delta p = 7°$, $\delta m = 12°$). If we include the Bukoban intrusives mean result from Piper (1972), the paleomagnetic pole cal-

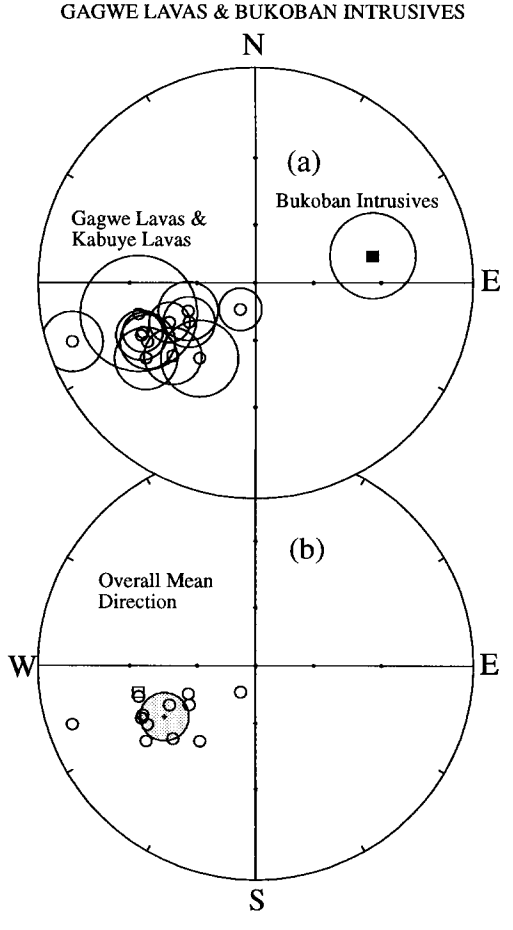


Fig. 7. (a) Site mean results and their α_{95} circles. Symbols are the same as in the stereoplots of Fig. 4. The square represents the direction obtained by Piper (1972) from the sixteen sites of Bukoban intrusives. (b) Mean direction and α_{95} circle about the mean direction obtained by reversing the Bukoban intrusives result (shown as a open square for visual reference).

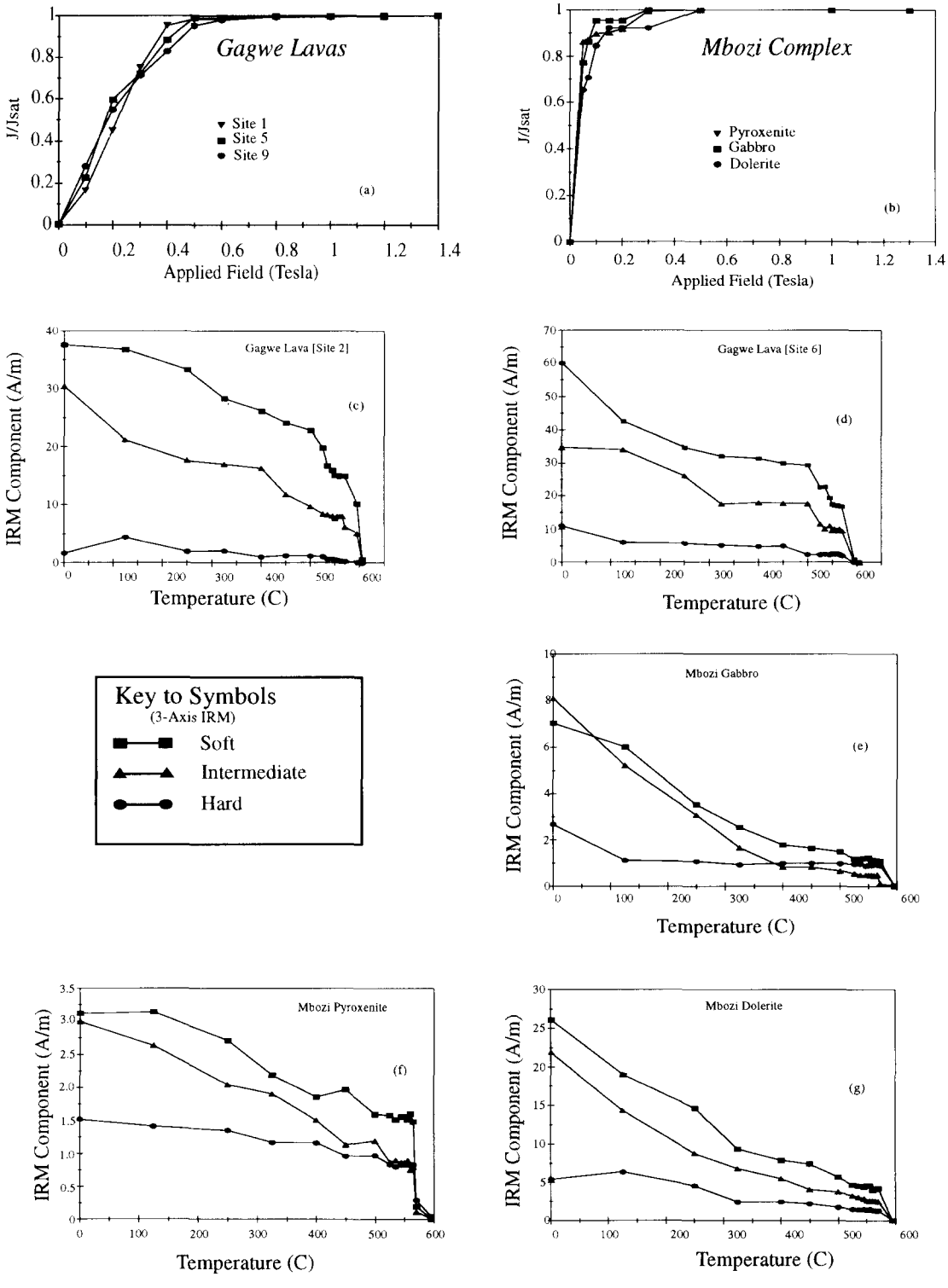


Fig. 8. Isothermal remanence acquisition (IRM) data for (a) some of the Gagwe lavas and (b) for samples taken from the Mbozi complex. Three-axis IRM demagnetization results from (c) a Gagwe lava sample at Site 2, (d) Gagwe lava at Site 6, (e) a gabbro in the Mbozi complex, (f) a pyroxenite in the Mbozi complex and (g) a dolerite in the Mbozi complex.

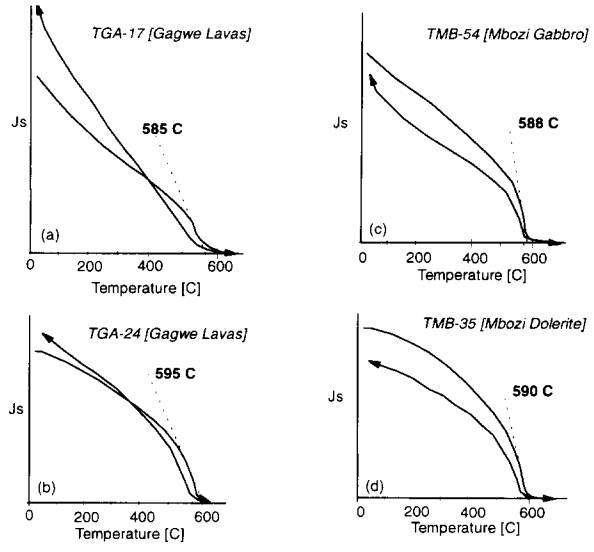


Fig. 9. Curie temperature runs showing heating and cooling curves for (a) sample TGA-17 of the Gagwe lavas with an estimated Curie temperature of 585°C, (b) sample TGA-24 of the Gagwe lavas with an estimated Curie temperature of 595°C, (c) sample TMB-54 of the Mbozi complex gabbros with an estimated Curie temperature of 588°C and (d) sample TMB-35 of the Mbozi complex dolerite with an estimated Curie temperature of 590°C.

culated from the mean of virtual geomagnetic poles (VGP's) is 25°S, 274°E ($K=17$, $A_{95}=10°$). The Bukoban intrusives result is reversed with respect to our extrusive rocks result (Fig. 7b) and yields a positive reversals test using the McFadden and McElhinny (1990) method with a classification of *C*. The Gagwe lavas pole is graded $Q=5$ according to the *Q*-factor of Van der Voo (1990).

The magnetization in the Bukoban lavas appears to be carried by fine-grained low-Ti magnetite based on the unblocking temperatures and coercivity spectra (Figs. 4–6). This is confirmed by isothermal remanence acquisition (IRM) studies (Fig. 8a) that may also show a small contribution from hematite or pyrrhotite, three-axis IRM demagnetization (Figs. 8c and 8d; Lowrie, 1990) and Curie temperature experiments (Figs. 9a and 9b).

The characteristic direction obtained from the lavas in our study is similar to that obtained by Piper (1972); however, the inclination obtained by us is 14° steeper than the mean inclination calculated in the original study ($I=-39°$ vs. $I=-25°$). Piper (1972) used blanket AF-demagnetization levels between 40 and 80 mT. As we noted above, most samples follow great-circle trajectories through a WSW shallow-up direction before reaching the stable endpoint direction. When we calculate a mean inclination from all our samples using the directional data at 40, 60 and 80 mT we observe the mean inclination to steepen from $-14°$ (40 mT) to $-25°$ (60 mT) and $-34°$ (80 mT). We therefore attribute the 14° difference in our inclination results to be due to incomplete cleaning of the samples in the original study.

3.2. The Mbozi complex

The intensity of the natural remanence magnetization (NRM) ranges between 10^{-4} and 6 A/m. Thermal and alternating field (AF) demagnetization on pilot specimens indicated thermal demagnetization to result in clearer definition of the magnetic components (Figs. 10 and 11). The NRM grouping ($D=352°$, $I=-24°$; Fig. 12a) in the Mbozi complex samples is consistent with the present Earth's co-axial dipole field ($D=0°$, $I=-18°$) at this location in southern Tanzania. Samples treated using alternating field demagnetization often showed erratic behavior at low fields and a tendency not to reach a stable endpoint direction (Figs. 10a and 10c). Most thermally treated samples from the Mbozi complex show two components of magnetization (Table 2). The first low-temperature (LT) component is typically removed between 0 and 550°C (Fig. 10b) and has a mean direction of $D=353°$, $I=-33°$ ($k=24$, $\alpha_{95}=10°$; Fig. 12b) that we attribute to a viscous overprint of recent origin. The characteristic, high-temperature (HT), direction resolved in 64 of the 66 samples collected from the ten sites is $D=320°$, $I=+16°$ ($k=32$, $\alpha_{95}=09°$; Figs. 12c–12e). The gabbro at Site 9 shows cryptic layering of gabbro and pyroxenite. The pyroxenite layer samples yield a direction that is approximately reversed with respect to the gabbros (see Table 2; Figs. 11a and 11c). The paleomagnetic pole calculated from the Mbozi complex characteristic mean direction is 46°N, 325°E ($\delta p=5°$, $\delta m=9°$). The Mbozi complex pole is graded at $Q=5$ according to the Van der Voo (1990) criteria.

The characteristic remanence in these rocks is carried by low-Ti magnetite as evidenced by the characteristic unblocking temperatures between 525 and 585°C (Fig.

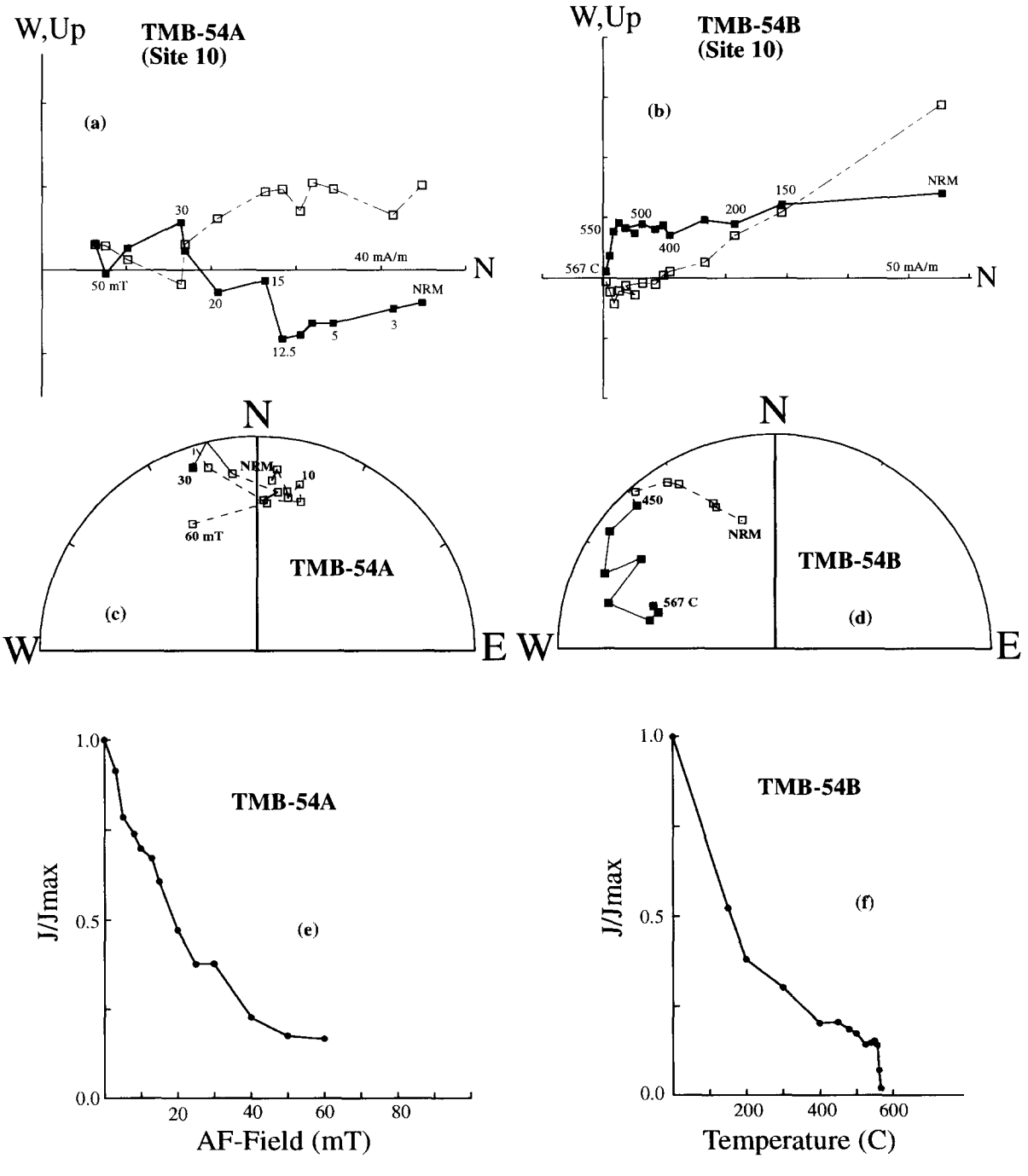


Fig. 10. Orthogonal vector diagrams, equal-area stereoplots and intensity decay plots for pilot samples of the Mbozi complex demonstrating the effectiveness of thermal demagnetization over alternating field demagnetization. Symbols the same as in Fig. 4.

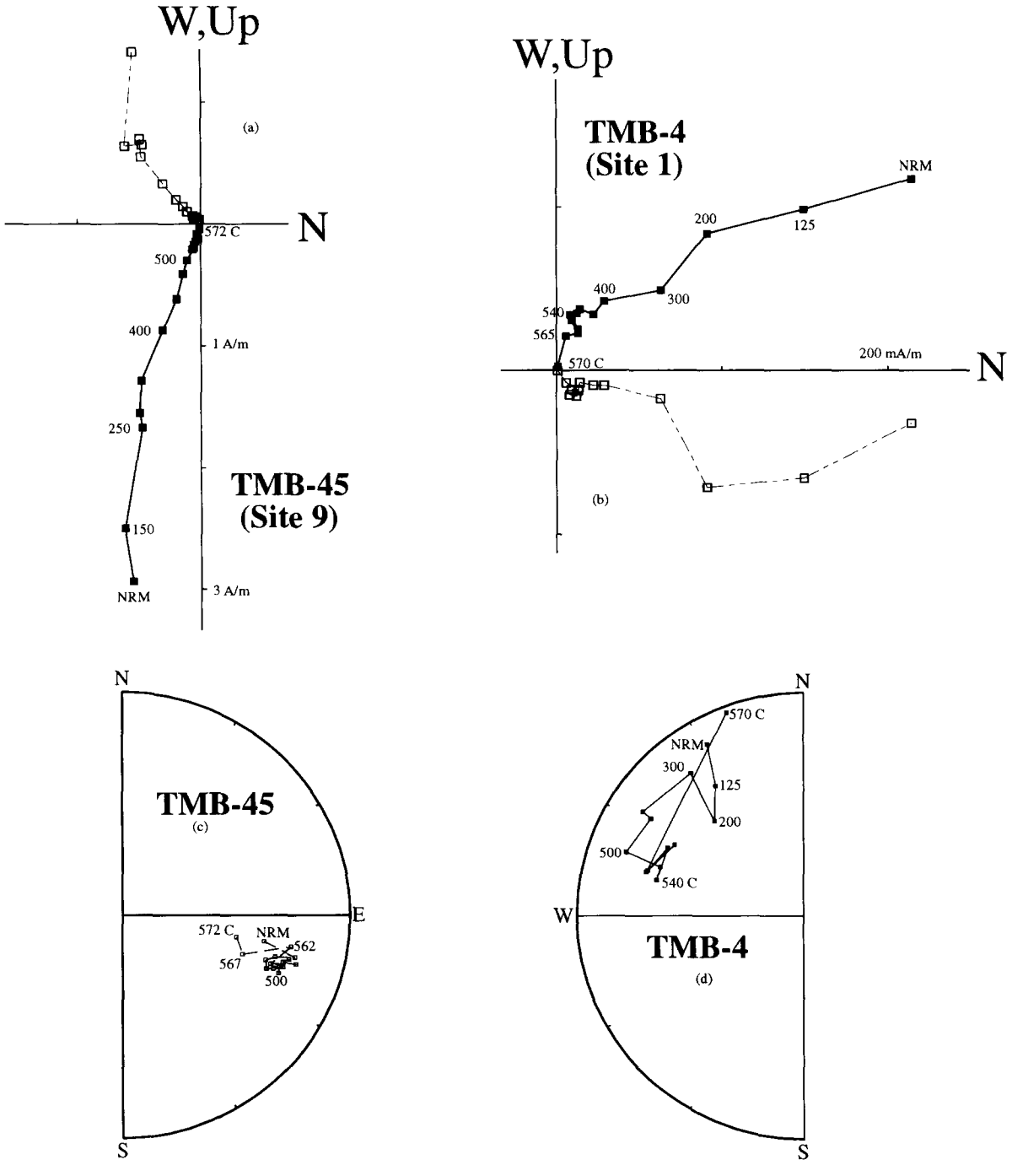


Fig. 11. (a, c) Orthogonal vector diagram and equal-area stereoplot for pyroxenite sample TMB-45 from Site 9 of the Mbozi complex that is approximately reversed from the gabbro/dolerite directions. (b, d) Orthogonal vector diagram and equal-area stereoplot for gabbro sample TMB-4 from Site 1 of the Mbozi complex. Symbols are the same as in Fig. 4.

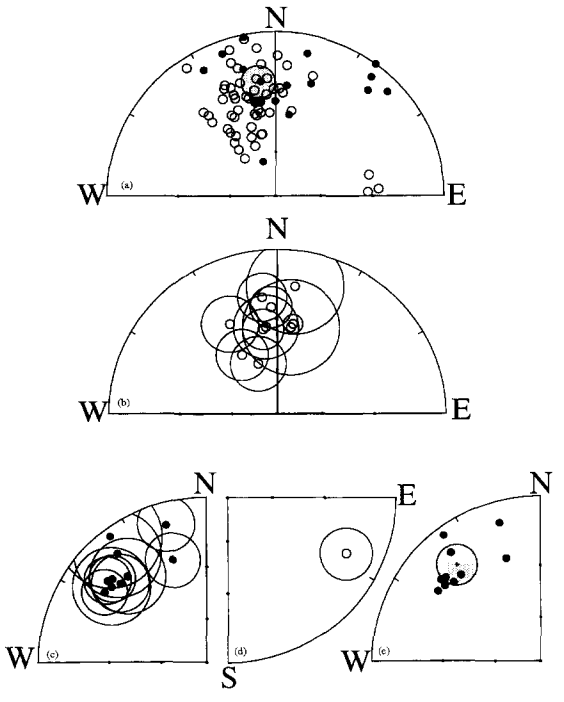


Fig. 12. (a) NRM directions measured from the Mbozi complex along with the mean direction and its α_{95} circle (shaded). (b) Low-temperature site mean directions and their α_{95} circles. (c) High-temperature site means from the Mbozi complex and their α_{95} circles. (d) opposite polarity mean from the pyroxenite at Site 9 and its α_{95} circle. (e) Overall mean direction and α_{95} circle obtained by reversing the direction in (d). Symbols are the same as in the stereoplots of Fig. 7.

10g), IRM experiments (Fig. 8b), three-axis IRM demagnetization (Figs. 8e–8g) and Curie temperature experiments (Figs. 9c and 9d).

The paleomagnetic pole obtained by us is very different from the paleomagnetic pole obtained by Piper (1975). The characteristic direction calculated in the original study is $D = 10°, I = +11°$ ($k = 19$, $\alpha_{95} = 18°$). Those samples were AF-treated and exhibited poor stability with no tendency to reach a stable endpoint (Piper, 1975). Individual site means in the original study have α_{95} values ranging from 16 to 58° with only one site having an α_{95} less than 27° (that site was not used in calculating the mean direction!). We attribute the different behavior of our results to the greater resolution power of thermal demagnetization compared with AF-demagnetization.

Table 2
Paleomagnetic results from the Mbozi complex

Site	n/N	Dec. (°)	Inc. (°)	k	α_{95} (°)	VGP lat.	VGP long.
1-LT	5/5	008	−14	15	20	–	–
1-HT	5/5	318	+20	21	17	44°N	327°E
2-LT	4/4	009	−34	6	28	–	–
2-HT	4/4	309	+18	21	17	36°N	321°E
3-LT	4/4	010	−32	62305		–	–
3-HT	4/4	312	+16	79	10	39°N	321°E
4-LT	7/7	353	−34	14	17	–	–
4-HT	7/7	313	+20	22	13	40°N	325°E
5-LT	6/6	330	−45	21	15	–	–
5-HT	5/6	321	+10	21	17	49°N	321°E
6-LT	6/6	340	−54	17	17	–	–
6-HT	5/6	323	+03	19	18	52°N	318°E
7-LT	6/6	357	−24	42	10	–	–
7-HT	6/6	342	+24	26	13	62°N	353°E
8-LT	6/6	353	−19	40	11	–	–
8-HT	6/6	344	+08	35	11	70°N	343°E
9-LT	9/9	333	−27	19	14	–	–
9-HT_r	5/5	321	+20	22	17	46°N	329°E
9-HT_n	4/4	116	−14	67	11	25°N	315°E
9-HT_c	9/9	310	+15	30	15	37°N	320°E
10-LT	13/13	351	−35	16	11	–	–
10-HT	13/13	305	+17	22	10	33°N	319°E
Total-LT	66/66	353	−33	24	10	–	–
Total-HT	64/66	320	+16	32	09	46°N	325°E
						$\delta p = 5°$	$\delta m = 9°$

LT = low-temperature component; HT = high-temperature component; HT_n = high-temperature normal polarity; HT_r = high-temperature reverse polarity; HT_c = high-temperature combined polarity; remaining abbreviations are identical to those found in Table 1. The Mbozi complex quarry is located at 9.25°S, 32.9°E.

4. Discussion

The paleomagnetic directions obtained from the Bukoban lavas (Gagwe and Kabuye), Bukoban intrusives (Piper, 1972) and Mbozi complex (this study) provide good reference poles for the Congo Craton at

810 and 743 Ma, respectively. These poles refine the currently available paleomagnetic database for the Congo Craton and allow us to test a variety of tectonic models proposed for the Congo Craton for the interval 850–505 Ma. Table 3 lists the paleomagnetic poles for the Congo Craton, Rio Plata Craton of South America, India, Australia, Baltica, West African Craton, Arabian–Nubian Shield and Laurentia with a Q-value $\geqslant 3$, ages known to within 6% and A_{95} (or α_{95}) less than 20°. The Kalahari Craton and East Antarctic Craton are the most notable absentees from this listing. There are paleomagnetic poles from the Kalahari Craton that fall within the age range 850–505 Ma; however, the age assignments for the multiple magnetization directions observed in the rocks preclude an unambiguous interpretation of the results (Van der Voo and Meert, 1991). Our analysis of the available paleomagnetic data seeks to simultaneously address issues regarding the breakup of Rodinia and formation of Gondwana. Assuming the validity of the Rodinia supercontinent, these two events must be intimately linked as the cratons that formed the margins of Rodinia later coalesced to form the Gondwana continent. How and when these events occurred must include answers to the following questions:

(a) When did Australia and East Antarctica rift from western Laurentia?

(b) How does the 800–650 Ma East African Orogen (Stern, 1994) relate to the assembly of Gondwana?

(c) What is the significance of the 550 Ma orogenic events throughout East Gondwana?

(d) When was Gondwana completely assembled?

Both Hoffman (1991) and Dalziel (1992) postulate that their Rodinia reconstructions are valid from roughly 1100 Ma until ~700 Ma at which time Australia and East Antarctica begin to break away from the western margin of Laurentia. However, there is a debate as to the exact timing of the onset of this rifting. Ross (1991) suggested that the passive margin sequence chronology in northwestern Laurentia is compatible with a breakup age of 800–700 Ma. Bond et al. (1984) examined passive margin sequences from around the globe and assuming that a supercontinental breakup was reflected in the thermal subsidence curves, argued for breakup of this supercontinent between 625 and 550 Ma. A subsequent analysis of the subsidence data by Kominz (1995) also argues for breakup along western Laurentia at sometime post-600 Ma. Meert and Van der Voo (1994) point out that a post-600 Ma breakup is permissible because a gap in the paleomagnetic database for Laurentia, Antarctica and Australia for the time period from 725 to 600 Ma precludes resolution of detailed continental motions during that interval (see listing in Table 3). We argue here that the later, post-600 Ma, passive margin development could arise from the breakup of Australo–Antarctica from western Laurentia.

In order to test the Rodinia configuration and determine the role of the Congo Craton in the reconstruction, we have rotated the 810–700 Ma poles from the available database into the reconstruction of Dalziel (1992) (Fig. 13). The Laurentian mean poles are taken from the analysis of Powell et al. (1993). There is general agreement between the ~720 Ma poles from India, Baltica and Laurentia as the confidence intervals just overlap. However, the 743 Ma Mbozi complex pole from the Congo Craton falls nearly 40° away from the Laurentian APWP after rotation and the 810 Ma Gagwe lavas pole is even more remote. This suggests that the Congo Craton was not a part of Rodinia as configured by Dalziel (1992). The 709 Ma pole from the Rio Plata Craton also falls well away from the Laurentian poles in the reconstruction and raises questions about the position of the Rio Plata Craton within Dalziel's (1992) configuration of Rodinia if the pole represents a primary remanence.

The timing for Gondwana assembly and an explanation for the 550 Ma orogenic events in East Gondwana can also be examined using the available paleomagnetic data. Stern (1994) has provided a detailed tectonic summary for the amalgamation of Gondwana in Neoproterozoic times. His model relates the East Africa Orogen (EAO), more commonly known as the Mozambique Belt, to the formation of Gondwana during the interval 800–650 Ma. According to this model, and that of Hoffman (1991) and Dalziel (1992), the Mozambique Belt results from the collision between fragments of West Gondwana and a united East Gondwana about a pole of rotation near the present-day tip of southern Africa (see Fig. 1a, southern tip of Kalahari block). Thus, in these models, Gondwana assembly is largely complete by 650 Ma. The models attribute the ubiquitous 550–480 Ma ages found throughout East Gondwana to a later disturbance of unknown significance. These models also conclude that the post-600 Ma passive margin formation in western Laurentia resulted from a thermal disturbance unrelated

Table 3
Listing of paleomagnetic poles from 810 to 505 Ma

Pole	Q-value[a]	Age ± error (Ma)	Pole lat.	Pole long.	A_{95} or α_{95} (°)	Rotated to Dalziel (1992)	Rotated to Gondwana	Reference
Congo Craton								
Gagwe lavas	5	810 ± 25	25°S	273°E	10	38°S, 183°E	fixed	this study
Mbozi complex	5	743 ± 30	46°N	325°E	09	40°S, 325°E	fixed	this study
Sinyai dolerite	5	547 ± 04	29°S	319°E	05	34°N, 310°E	fixed	Meert et al. (1994b)
Ntonya ring	4	522 ± 13	28°N	345°E	02	28°S, 300°E	fixed	Briden et al. (1993)
Rio Plata Craton								
La Tinta Fm.	5	709 ± 24	80°S	301°E	05	14°N, 254°E	51°S, 63°E	Valencio et al. (1980)
India								
Malani rhyolites	5	729 ± 10	81°S	224°E	10	16°S, 342°E	42°S, 279°E	Klootwijk (1975)
Bhander and Rewa s.str.	3	~550	51°N	217°E	11	29°N, 097°E	23°N, 153°E	McElhinny et al. (1978)
Purple sandstone	3	530 ± 15	28°N	212°E	12	21°N, 087°E 05°N,	168°E	McElhinny (1970)
Jutana Fm.—Pakistan	4	530 ± 12	21°N	231°E	11	16°N, 065°E	14°N, 185°E	Klootwijk et al. (1986)
Salt pseudomorph's	4	523 ± 24	27°N	214°E	5	33°N, 069°E	06°N, 170°E	Wensink (1972)
central Australia								
Mean pole	6	550 ± 15	45°S	346°E	13	29°S, 265°E	12°S, 343°E	Means calculated from data
Mean pole	6	530 ± 15	36°S	005°E	18	35°S, 247°E	02°N, 351°E	compilation of Li and
Mean pole	6	510 ± 15	26°S	022°E	18	37°S, 225°E	17°N, 001°E	Powell (1993)
Baltica								
Mean pole	4	750 ± 15	27°S	021°E	09	20°S, 002°E	N/A	Torsvik et al. (1995)
Mean pole	4	630 ± 15	25°N	051°E	15	33°N, 031°E	N/A	Torsvik et al. (1995)
Mean pole	3	580 ± 15	69°N	093°E	11	74°N, 092°E	N/A	Torsvik et al. (1995)
Mean pole	4	565 ± 15	57°N	143°E	15	55°N, 134°E	N/A	Torsvik et al. (1992)
Mean pole	3	553 ± 15	38°N	134°E	12	38°N, 120°E	N/A	Torsvik et al. (1992)
West African Craton								
Adma diorite	3	608 ± 18	33°N	345°E	9	10°N, 005°E	fixed	Morel (1981)
Arabian–Nubian Shield								
Nabati batholith	3	587 ± 18	68°N	314°E	19	58°S, 345°E	fixed	Saradeth et al. (1989)
Laurentia								
Mean pole	4	810 ± 40	11°N	324°E	15	fixed	N/A	Powell et al. (1993)
Mean pole	6	780 ± 05	06°N	318°E	06	fixed	N/A	Powell et al. (1993)
Mean pole	6	725 ± 15	06°S	342°E	15	fixed	N/A	Powell et al. (1993)
Mean pole	6	580 ± 20	44°N	304°E	9	fixed	N/A	Meert et al. (1994c)
Mean pole	4	550 ± 15	13°N	345°E	15	fixed	N/A	Meert et al. (1994c)
Mean pole	6	505 ± 15	03°S	344°E	12	fixed	N/A	Meert et al. (1994c)

Euler poles for Dalziel (1992): Congo 2.8°N, 41.3°W, −167.6°; Rio Plata 1.9°N, 22.3°W, −97.7°; India 53.1°N, 145.1°E, +167.9°; Australia 28.9°N, 126.1°E, +132.1°; Baltica 68.3°N, 127.7°E, −22.4°; West Africa 18.3°N, 9°W, −138.6°; Arabian–Nubian Shield 2.8°N, 41.3°W, −167.6°. Dalziel reconstruction holds Laurentia fixed. Gondwana Euler poles from De Wit et al. (1988, Africa fixed); India 29.6°N, 36.1°E, −56.8°; Australia 24.6°S, 60.4°W, +49.9°; Rio Plata 45.5°N, 32.2°W, +57.5°. All 600 Ma and younger poles are south poles, older poles have polarity ambiguity. The mean 580 Ma pole for North America is recalculated from Meert et al. (1994c) which contained an error in the position of the Catoctin pole. The recalculated position for the Catoctin pole is 43°N, 298°E.

[a] Q-value for paleomagnetic pole according to criteria set forth by Van der Voo (1990).

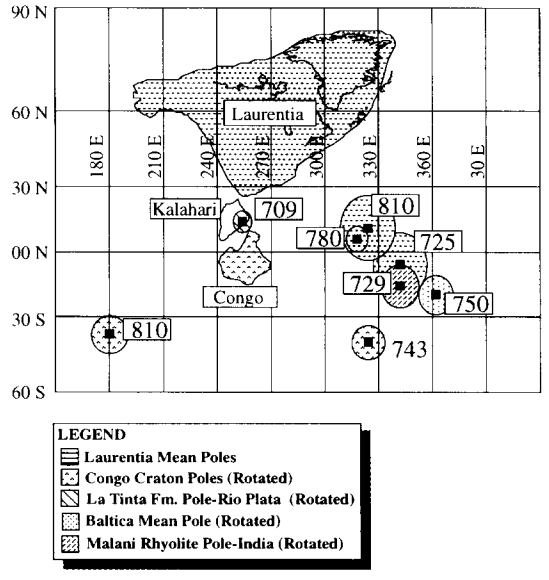


Fig. 13. Paleomagnetic poles from Table 3 in the age range from 810 to 709 Ma rotated into the configuration of Dalziel (1992). Laurentian poles are fixed. Note the coincidence of the Baltica and India poles with Laurentian poles and the disparity between the Congo Craton, Rio Plata Craton and Laurentian poles.

to the rifting of Australo–Antarctica. Below, we suggest an alternative scenario for Gondwana assembly. Fig. 14 shows the available 810–510 Ma paleomagnetic poles from the Gondwana cratons rotated into African coordinates. Although the 810–585 Ma poles are few and scattered, preventing detailed conclusions regarding Gondwana assembly, this scatter may actually reflect the fact that Gondwana assembly was not yet complete. On the other hand, the 550–510 Ma poles form a reasonably coherent swathe from 29°S, 319°E, to northwestern Africa (Fig. 14b). This swathe supports a united Gondwana by at least 550 Ma.

There appear to be at least two separate tectonic events within East Africa that were previously ascribed to a rather ill-defined "Pan-African" event. The Pan-African episode refers to polyphase tectonometamorphic events that occurred between 800 and 450 Ma within Gondwana. Recent geochronologic data indicate the presence of distinct orogenic events within Gondwana rather than a single, prolonged, Pan-African event (Stern, 1994 and references therein). The East Africa Orogen (800–650 Ma; Stern, 1994) represents a separate tectonic event within the Pan-African of central Gondwana. If the East Africa Orogen is responsible for the assembly of greater Gondwana it will find little support from the available paleomagnetic data (Fig. 14a). This may be an artifact of an incomplete paleomagnetic record or it may reflect that only the later 550 Ma orogenic events recognized in East Gondwana represent the final suturing of the Gondwana continent. However, if the 550 Ma event represents the final suturing of a united East Gondwana with the formerly fragmented West Gondwana blocks, what, if anything, collided earlier with the Congo Craton to form the East Africa Orogen? We tentatively suggest the abandonment of a united East Gondwana during late Neoproterozoic times and offer the alternative of polyphase accretion of terranes between 800 and 550 Ma to form greater Gondwana. Our model proposes that the East

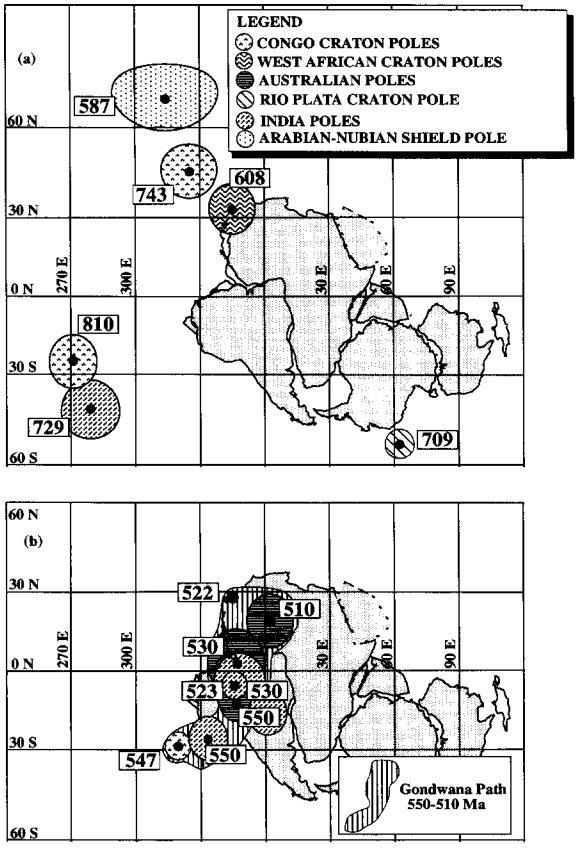


Fig. 14. (a) Gondwana paleomagnetic poles from Table 3 in the age range 810–585 Ma rotated to the Gondwana configuration of De Wit et al. (1988) with the African poles fixed. (b) Gondwana paleomagnetic poles (south poles) for the interval 550–510 Ma rotated to Africa coordinates.

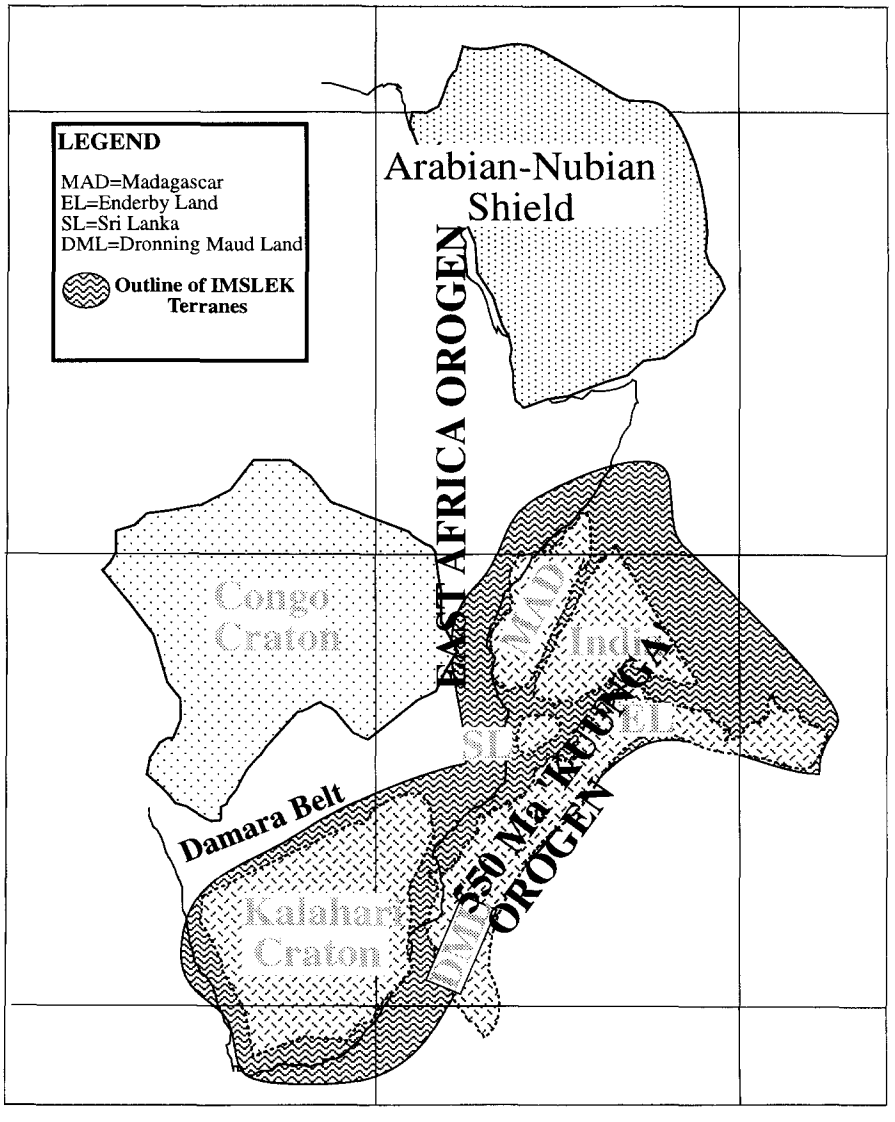


Fig. 15. Outline of the IMSLEK terranes that collide with the Congo Craton and Arabian–Nubian Shield (ANS) to form the Mozambique Belt between 800 and 650 Ma. This collision is followed 100–250 m.y. later (550 Ma) by the collision of Australia–Antarctica with IMSLEK–Congo–Arabian/Nubian Shield during the Kuunga Orogeny with possible sutures now exposed in Sri Lanka and East Antarctica.

Africa Orogen formed as the result of a 800–650 Ma collision between the IMSLEK terranes (*I*ndia, *M*adagascar, *S*ri *L*anka, a strip of the East Antarctic Craton including *E*nderby Land and Dronning Maud Land, and the *K*alahari Craton; Fig. 15) with the Congo Craton and Arabian–Nubian Shield (ANS) closing the northern part of the Mozambique Ocean and an ocean that disappeared in the Damara Belt. Our model then ascribes the ubiquitous 550 Ma ages within East Gondwana to the collision of Australo–Antarctica with the combined IMSLEK–Congo–ANS continent in a final Gondwana assembly. Our model is not inconsistent with the paleomagnetic observations cited above although more paleomagnetic data from the Gondwana blocks during the interval 800–550 Ma will ultimately provide a clearer test of the model. Our model does

require the existence of a suture running through a portion of East Antarctica, Sri Lanka and southeastern India with an approximate 550 Ma age. Indeed, such orogenic belts and sutures have been proposed by Shiriashi et al. (1994) in Enderby Land (Antarctica) and in India and Sri Lanka by Kröner (1993).

5. Conclusions

A paleomagnetic investigation of the 810±25 Ma Gagwe–Kabuye lavas of western Tanzania and eastern Burundi yields a paleomagnetic pole at 25°S, 273°E ($\delta p = 7°$, $\delta m = 12°$). The 743±30 Ma Mbozi gabbro-syenite complex of southern Tanzania yields a paleomagnetic pole at 46°N, 325°E ($\delta p = 5°$, $\delta m = 9°$). These two paleomagnetic poles augment the database for the Congo Craton in East Africa and allow us to test a variety of proposed tectonic models for the region in Neoproterozoic times. Collectively, these paleomagnetic data suggest that the Congo Craton may not have been a part of the supercontinent Rodinia in the configuration envisioned by Dalziel (1992). Paleomagnetic poles from the Gondwana cratonic blocks are compatible with, though do not themselves prove, a Gondwana assembly through two separate orogenic events. The East Africa Orogen represents the collision between the IMSLEK terranes as defined in this paper and the Congo Craton and Arabian–Nubian Shield (ANS) between 800 and 650 Ma. A later collision between Australo–Antarctica and IMSLEK–Congo–ANS then occurred around 550 Ma. Our proposal requires the separation of East Gondwana into two or more fragments and we attribute the ubiquitous 550 Ma ages within east and central Gondwana to a separate orogenic event which we call the Kuunga Orogen after the Swahili word Kuungana, meaning "to join or unite". A clear test of this model requires sutures of 550 Ma age within East Gondwana such as those documented by Shiriashi et al. (1994) and Kröner (1993) along with additional paleomagnetic data from the Gondwana blocks.

Acknowledgements

We wish to thank the Division of Mines and Geology in Dodoma, Tanzania for their assistance in all phases of this research. We also wish to thank Makenya Maboko for his hospitality during our stay in Tanzania, the University of Dar es Salaam for providing a rental vehicle for part of our field work and to Rob Hargraves of Princeton University for supplying us with paleomagnetic samples from the Kabuye lavas in Burundi. The manuscript was greatly improved through early reviews by Trond Torsvik and Ben van der Pluijm and editorial reviews by Chris Klootwijk, Alfred Kröner, Z.X. Li and Lyle Harris. We also thank Harald Walderhaug at the University of Bergen, Norway for providing us with the results of Curie temperature runs. This research was supported by the Division of Earth Sciences, National Science Foundation, grant EAR 92-05815.

References

Abdelsam, M.G. and Stern, R.J., 1993. Tectonic evolution of the Nakasib suture, Red Sea Hills, Sudan: evidence for a Wilson cycle in the Late Precambrian of the Arabian–Nubian shield. J. Geol. Soc. London, 150: 393–404.

Bond, G.C., Nickeson, P.A. and Kominz, M.A., 1984. Breakup of a supercontinent between 625 Ma and 555 Ma: new evidence and implications for continental histories. Earth Planet. Sci. Lett., 70: 325–345.

Briden, J.C., McClelland, E. and Rex, D.C., 1993. Proving the age of a paleomagnetic pole: the case of the Ntonya ring structure, Malawi. J. Geophys. Res., 98: 1743–1749.

Brock, A., 1968. Metasomatic and intrusive nepheline-bearing rocks from the Mbozi syenite-gabbro complex, southwestern Tanzania. Can. J. Earth Sci., 5: 387–419.

Burke, K. and Dewey, J.F., 1972. Orogeny in Africa. In: T.F.J. Dessauvigie and A.J. Whiteman (Editors), Orogeny in Africa. Univ. of Ibadan, Ibadan, pp. 583–608.

Dalziel, I.W.D., 1991. Pacific margin of Laurentia and East Antarctica/Australia as a conjugate rift pair: evidence and implications for an Eocambrian supercontinent. Geology, 19: 586–601.

Dalziel, I.W.D., 1992. On the organization of American plates in the Neoproterozoic and the breakout of Laurentia. GSA Today, 2(11): 237–241.

De Paepe, P., Tack, L., Moens, L. and Van de Velde, P., 1991. The basic magmatism of the Upper Proterozoic in south-east Burundi. Mus. R. Afr. Cent. Tervuren Annu. Rep., 1989–1990: 85–104.

De Wit, M., Jeffrey, M., Bergh, H. and Nicolaysen, L., 1988. Geologic map of sectors of Gondwana reconstructed to their disposition at 150 Ma (1:10,000,000). American Association of Petroleum Geologists, 1988.

Hanson, R.E., Wardlaw, M.S., Wilson, T.J. and Mwale, G., 1993. U–Pb zircon ages from the Hook granite massif and Mwembeshi dislocation: constraints on Pan-African deformation, plutonism, and transcurrent shearing in central Zambia. Precambrian Res., 63: 189–209.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwanaland inside-out? Science, 252: 1409–1412.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J.R. Astron. Soc., 67: 699–718.

Klootwijk, C.T., 1975. A note on the paleomagnetism of the late Precambrian Malani rhyolites near Jodhpur—India. J. Geophys., 41: 189–200.

Klootwijk, C.T., Nazirullah, R. and De Jong, K.A., 1986. Paleomagnetic constraints on formation of the Mianwali reentrant, Trans-Indus and western Salt Range, Pakistan. Earth Planet. Sci. Lett., 80: 394–414.

Kominz, M.A., 1995. Thermally subsiding basins and the effect of sediment; with application to the Cambro–Ordovician Great Basin sequence, western U.S. Basin Res. (in press).

Kröner, A., 1993. The Pan African Belt of northeastern and Eastern Africa, Madagascar, southern India, Sri Lanka and East Antarctica: terrane amalgamation during the formation of the Gondwana supercontinent. In: Geoscientific Research in Northeast Africa. Balkema, Rotterdam, pp. 3–9 (abstract).

Li, Z.X. and Powell, C. McA., 1993. Late Proterozoic to Early Paleozoic paleomagnetism and the formation of Gondwanaland. In: R.H. Findlay et al. (Editors), Gondwana 8: Assembly, Evolution and Dispersal. Balkema, Rotterdam, pp. 9–21.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett., 17: 159–162.

McElhinny, M.W., 1970. Paleomagnetism of the Cambrian Purple sandstone from the Salt Range, west Pakistan. Earth Planet. Sci. Lett., 9: 149–156.

McElhinny, M.W., Cowley, J.A. and Edwards, D., 1978. Paleomagnetism of some rocks from peninsular India and Kashmir. Tectonophysics, 50: 41–54.

McFadden, P.L. and McElhinny, M.W., 1990. Classification of the reversal test in paleomagnetism. Geophys. J. Int., 103: 725–729.

Meert, J.G. and Van der Voo, R., 1994. The Neoproterozoic [1000–540 Ma] glacial intervals: no more snowball Earth? Earth Planet. Sci. Lett., 125: 1–13.

Meert, J.G., Hargraves, R.B., Van der Voo, R., Hall, C.M. and Halliday, A.N., 1994a. Paleomagnetic and $^{40}Ar/^{39}Ar$ studies of Late Kibaran intrusives in Burundi, East Africa: implications for Late Proterozoic supercontinents. J. Geol., 102: 621–637.

Meert, J.G., Uranga, C., Van der Voo, R., Ayub, S. and Munyua, R., 1994b. Middle and Late Proterozoic paleomagnetic results from the Congo craton, East Africa. EOS, 75(16): 128.

Meert, J.G., Van der Voo, R. and Payne, T., 1994c. Paleomagnetism of the Catoctin volcanic province: a new Vendian–Cambrian apparent polar wander path for North America. J. Geophys. Res., 99: 4625–4641.

Moores, E.M., 1991, Southwest U.S.–East Antarctica (SWEAT) connection: a hypothesis. Geology, 19: 425–428.

Morel, P., 1981. Paleomagnetism of a Pan-African diorite: a late Precambrian pole for western Africa. Geophys. J.R. Astron. Soc., 65: 493–503.

Piper, J.D.A., 1972. A paleomagnetic study of the Bukoban System, Tanzania. Geophys. J.R. Astron. Soc., 28: 111–127.

Piper, J.D.A., 1975. Paleomagnetic correlations of Precambrian formations of east-Central Africa and their tectonic implications. Tectonophysics, 26: 135–161.

Powell, C. McA., Li, Z.X., McElhinny, M.W., Meert, J.G. and Park, J.K., 1993. Paleomagnetic constraints on timing of the Neoproterozoic breakup of Rodinia and Cambrian formation of Gondwana. Geology, 21: 889–892.

Ross, G.M., 1991. Tectonic setting of the Windemere Supergroup: the continental perspective. Geology, 19: 1125–1128.

Saradeth, S., Soffel, P., Horn, P., Muller-Sohnius, D. and Schult, A., 1989. Upper Proterozoic and Phanerozoic pole positions and potassium–argon (K–Ar) ages from the East Sahara craton. Geophys. J. Int., 97: 209–221.

Schandelmeier, H., Abdel, E.M., Rahman, E., Wipfler, D., Küster, D., Utke, A. and Matheis, G., 1994. Late Proterozoic magmatism in the Nakasib suture, Red Sea Hills, Sudan. J. Geol. Soc. London, 151: 485–497.

Shiriashi, K., Ellis, D.J., Hiroi, Y., Fanning, C.M., Motoyoshi, Y. and Nakai, Y., 1994. Cambrian orogenic belt in East Antarctica and Sri Lanka: implications for Gondwana assembly. J. Geol., 102: 47–65.

Stern, R.J., 1994. Arc assembly and continental collision in the Neoproterozoic East African orogen: implications for the consolidation of Gondwanaland. Annu. Rev. Earth Planet. Sci., 22: 319–351.

Tack, L., Sindayihebura, A. and Cimpaye, D., 1992. The Nkoma (SE Burundi): an episodically reactivated lower Burundian (Middle Proterozoic) siliciclastic sequence, locally overlain by a Malagarasian (Upper Proterozoic) sedimentary breccia. I.G.C.P. Newsl., 255(4): 31–43.

Torsvik, T.H., 1992. IAPD Interactive analysis of palaeomagnetic data. NGU, Trondheim, 51 pp.

Torsvik, T.H., Smethurst, M.A., Van der Voo, R., Trench, A., Abrahamsen, N. and Halvorsen, E., 1992. Baltica. A synopsis of Vendian–Permian paleomagnetic data and their paleotectonic implications. Earth Sci. Rev., 3: 133–152.

Torsvik, T.H., Roberts, D. and Siedlecka, A., 1995. Riphean and Vendian paleomagnetic data from northeast Russia and northern Norway: review and supplementary data. Nor. Geol. Tidskkr. Spec. Ed. (in press).

Valencio, D.A., Sinito, A.M. and Vilas, J.F., 1980. Paleomagnetism of Upper Precambrian rocks of the La Tinta Formation, Argentina. Geophys. J.R. Astron. Soc., 62: 563–575.

Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics, 184: 1–9.

Van der Voo, R. and Meert, J.G., 1991. Late Proterozoic paleomagnetism and tectonic models: a critical appraisal. Precambrian Res., 53: 149–163.

Wensink, H., 1972. The paleomagnetism of the Salt Range pseudomorph beds of Middle Cambrian age from the Salt Range, West Pakistan. Earth Planet. Sci. Lett., 16: 189–194.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: D.W. Collinson, K.M. Creer and S.K. Runcorn (Editors), Methods in Paleomagnetism. Elsevier, New York, NY, pp. 254–286.